

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 12

REVIEW Lecture 11:

- **Finite Differences based Polynomial approximations**

- Obtain polynomial (in general un-equally spaced), then differentiate as needed
 - Newton's interpolating polynomial formulas

Triangular Family of Polynomials
(case of Equidistant Sampling,
similar if not equidistant)

$$f(x) = f_0 + \frac{\Delta f_0}{h}(x - x_0) + \frac{\Delta^2 f_0}{2!h^2}(x - x_0)(x - x_1) + \cdots \\ + \frac{\Delta^n f_0}{n!h^n}(x - x_0)(x - x_1) \cdots (x - x_{n-1}) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0) \cdots (x - x_n)$$

- Lagrange polynomial
(Reformulation of Newton's polynomial)

$$f(x) = \sum_{k=0}^n L_k(x) f(x_k) \quad \text{with} \quad L_k(x) = \prod_{j=0, j \neq k}^n \frac{x - x_j}{x_k - x_j}$$

- Hermite Polynomials and Compact/Pade's Difference schemes

(Use the values of the function and
its derivative(s) at nodes)

$$\sum_{i=-r}^s b_i \left(\frac{\partial^m u}{\partial x^m} \right)_{j+i} - \sum_{i=-p}^q a_i u_{j+i} = \tau_{\Delta x}$$

- **Finite Difference: Boundary conditions**

- Different approx. at and near the boundary => impacts global order of accuracy and linear system to be solved

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REVIEW Lecture 11:

- **Finite Difference: Boundary conditions**
 - Different approx. at and near the boundary => impacts linear system to be solved
- **Finite-Differences on Non-Uniform Grids and Uniform Errors: 1-D**
 - If non-uniform grid is refined, error due to the 1st order term decreases faster than that of 2nd order term
 - Convergence becomes asymptotically 2nd order (1st order term cancels)
- **Grid-Refinement and Error estimation**
 - Estimation of the order of convergence and of the discretization error
 - Richardson's extrapolation and iterative improvements using Roomberg's algorithm

FINITE DIFFERENCES – Outline for Today

-Differences on Non-Uniform Grids and Uniform Errors: 1-D

- Grid Refinement and Error Estimation
- Fourier Analysis and Error Analysis
 - Differentiation, definition and smoothness of solution for $\neq$ order of spatial operators
- Stability
 - Heuristic Method
 - Energy Method
 - Von Neumann Method (Introduction) : 1st order linear convection/wave eqn.
- Hyperbolic PDEs and Stability
 - Example: 2nd order wave equation and waves on a string
 - Effective numerical wave numbers and dispersion
 - CFL condition:
 - Definition
 - Examples: 1st order linear convection/wave eqn., 2nd order wave eqn., other FD schemes
 - Von Neumann examples: 1st order linear convection/wave eqn.
 - Tables of schemes for 1st order linear convection/wave eqn.

References and Reading Assignments

- Lapidus and Pinder, 1982: Numerical solutions of PDEs in Science and Engineering. Section 4.5 on “Stability”.
- Chapter 3 on “Finite Difference Methods” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”
- Chapter 3 on “Finite Difference Approximations” of “H. Lomax, T. H. Pulliam, D.W. Zingg, *Fundamentals of Computational Fluid Dynamics (Scientific Computation)*. Springer, 2003”
- Chapter 29 and 30 on “Finite Difference: Elliptic and Parabolic equations” of “Chapra and Canale, Numerical Methods for Engineers, 2014/2010/2006.”

Grid-Refinement and Error estimation

- We found that for a convergent scheme, the discretization error ε is of the form: $\varepsilon = \alpha O(\Delta x^p) + R$ (recall: $\phi = \hat{\phi} + \varepsilon$, $\mathcal{L}(\phi) = 0$, $\hat{\mathcal{L}}_{\Delta x}(\hat{\phi}) = 0$)
where R is the remainder
- The degree of accuracy and discretization error can be estimated between solutions obtained on systematically refined/coarsened grids
 - True solution u can be expressed either as:
$$\begin{cases} u = u_{\Delta x} + \beta \Delta x^p + R \\ u = u_{2\Delta x} + \beta' (2\Delta x)^p + R' \end{cases}$$
 - Thus, the exponent p can be estimated:
$$p \approx \log \left(\frac{u_{2\Delta x} - u_{4\Delta x}}{u_{\Delta x} - u_{2\Delta x}} \right) / \log 2$$

(need to eliminate u and then need 2 eqns. to eliminate both Δx and p , hence $u_{4\Delta x}$)
 - The discretization error on the grid Δx can be estimated by:
$$\varepsilon_{\Delta x} \approx \frac{u_{\Delta x} - u_{2\Delta x}}{2^p - 1}$$
 - Good idea: estimate p to check code. Is it equal to what it is supposed to be?
 - When solutions on several grids are available, an approximation of higher accuracy can be obtained from the remainder: Richardson Extrapolation!

Richardson Extrapolation and Romberg Integration

Richardson Extrapolation: method to obtain a third improved estimate of an integral based on two other estimates

Consider:

$$I = I(h) + E(h)$$

For two different grid space h_1 and h_2 :

$$I(h_1) + E(h_1) = I(h_2) + E(h_2)$$

Trapezoidal Rule:

$$E(h) = -\frac{b-a}{12}h^2 \hat{f}''$$

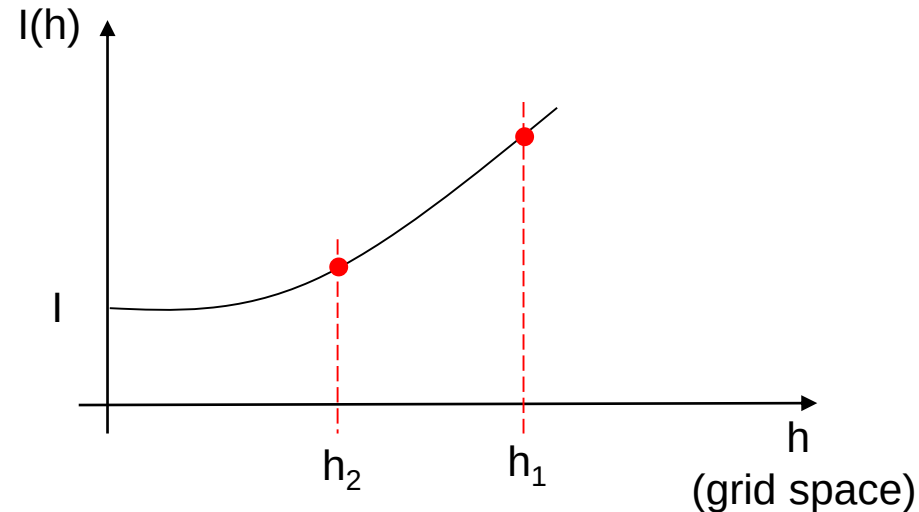
$$\Rightarrow E(h_1) \approx E(h_2) \left(\frac{h_1}{h_2}\right)^2$$

$$I(h_1) + E(h_2) \left(\frac{h_1}{h_2}\right)^2 \simeq I(h_2) + E(h_2)$$

$$E(h_2) \simeq \frac{I(h_1) - I(h_2)}{1 - (h_1/h_2)^2}$$

Richardson Extrapolation:

$$I = I(h_2) + \frac{I(h_2) - I(h_1)}{(h_1/h_2)^2 - 1} + O(h^4)$$



Example

Assume: $h_2 = h_1/2$

$$I = I(h_2) + \frac{I(h_2) - I(h_1)}{(2^2 - 1)} + O(h^4)$$

$$= \frac{4}{3}I(h_2) - \frac{1}{3}I(h_1) + O(h^4)$$

From two $O(h^2)$,
we get an $O(h^4)$!

Romberg's Integration:

Iterative application of Richardson's extrapolation

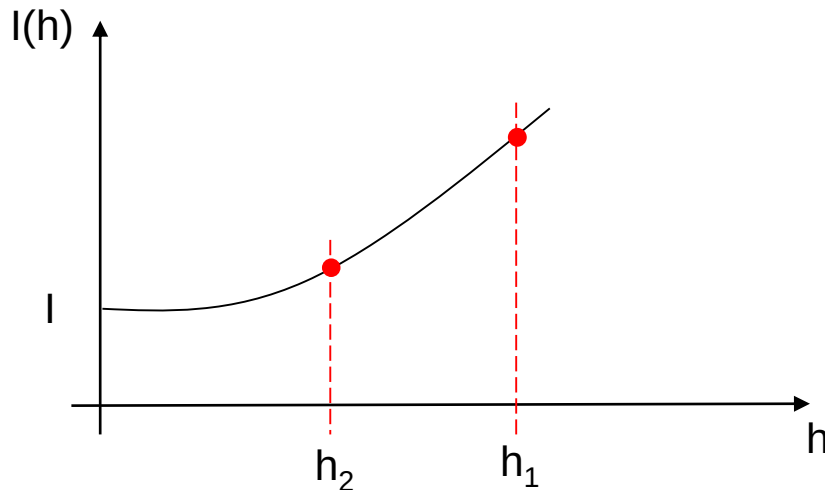
Romberg Integration Algorithm, for any order k

$$I_{j,k} \simeq \frac{4^{k-1} I_{j+1,k-1} - I_{j,k-1}}{4^{k-1} - 1}$$

For Order 2 (case of previous slide):

$$k = 2, j = 1$$

$$I_{1,2} \simeq \frac{4I_{2,1} - I_{1,1}}{3}$$



k Increasing order

1: $O(h^2)$ 2: $O(h^4)$ 3: $O(h^6)$ 4: $O(h^8)$

a. 0.172800 → 1.367467
 1.068800 →

b., 0.172800 1.367467 → 1.640533
 1.068800 → 1.623467 →
 1.484800 →

c. 0.172800 1.367467 1.640533 → 1.640533
 1.068800 1.623467 → 1.640533 →
 1.484800 → 1.639467 →
 1.600800 →

j ↓

Increasing resolution

Romberg's Differentiation:

Iterative application of Richardson's extrapolation

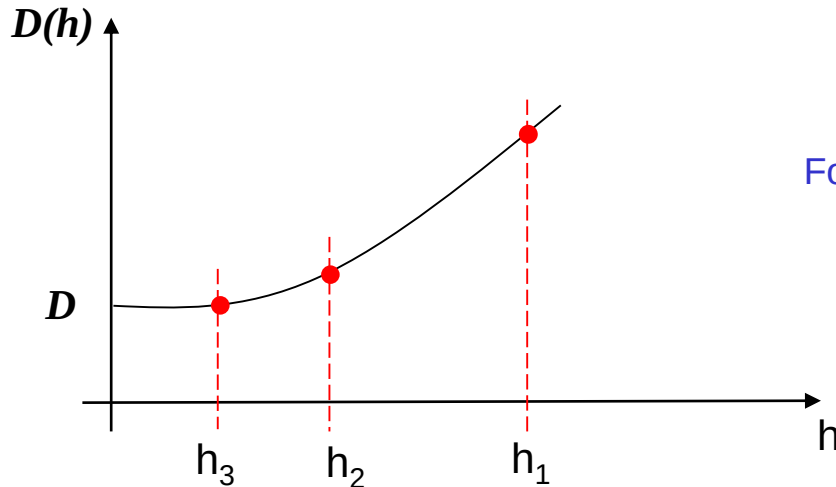
'Romberg' Differentiation Algorithm, for any order k

$$D_{j,k} \simeq \frac{4^{k-1} D_{j+1,k-1} - D_{j,k-1}}{4^{k-1} - 1}$$

For Order 2 (as previous slide, but for differentiation):

$$k = 2, j = 1$$

$$D_{1,2} \simeq \frac{4D_{2,1} - D_{1,1}}{3}$$



k → Increasing order

1: $O(h^2)$ 2: $O(h^4)$ 3: $O(h^6)$ 4: $O(h^8)$

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Increasing resolution j ↓

Fourier (Error) Analysis: Definitions

- Leading error terms and discretization error estimates can be complemented by a Fourier error analysis
- Fourier decomposition:
 - Any arbitrary periodic function can be decomposed into its Fourier components:

$$f(x) = \sum_{k=-\infty}^{\infty} f_k e^{ikx} \quad (k \text{ integer, wavenumber})$$

$$\int_0^{2\pi} e^{ikx} e^{-imx} dx = 2\pi \delta_{km} \quad (\text{orthogonality property})$$

Using the orthog. property,
taking the integral/FT of $f(x)$:

$$f_k = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-ikx} dx$$

- Note: rate at which $|f_k|$ with $|k|$ decays determine smoothness of $f(x)$
 - Examples drawn in lecture: $\sin(x)$, Gaussian $\exp(-\pi x^2)$, multi-frequency functions, etc

Fourier (Error) Analysis: Differentiations

- Consider the decompositions:

$$f(x) = \sum_{k=-\infty}^{\infty} f_k e^{ikx} \quad \text{or} \quad f(x,t) = \sum_{k=-\infty}^{\infty} f_k(t) e^{ikx}$$

- Taking spatial derivatives gives: $\frac{\partial^n f}{\partial x^n} = \sum_{k=-\infty}^{\infty} f_k(t) (ik)^n e^{ikx}$

- Taking temporal derivatives gives: $\frac{\partial^r f}{\partial t^r} = \sum_{k=-\infty}^{\infty} \frac{d^r f_k(t)}{dt^r} e^{ikx}$

- Hence, in particular, for even or odd spatial derivatives:

$$n = 2q \quad \Rightarrow \quad (ik)^n = (-1)^q k^{2q} \quad (\text{real})$$

$$n = 2q - 1 \quad \Rightarrow \quad (ik)^n = -i (-1)^q k^{2q-1} \quad (\text{imaginary})$$

Fourier (Error) Analysis: Generic equation

- Consider the generic PDE:

$$\frac{\partial f}{\partial t} = \frac{\partial^n f}{\partial x^n}$$

- Fourier Analysis: $f(x, t) = \sum_{k=-\infty}^{\infty} f_k(t) e^{ikx}$

- Hence:
$$\sum_{k=-\infty}^{\infty} \frac{df_k(t)}{dt} e^{ikx} = \sum_{k=-\infty}^{\infty} f_k(t) (ik)^n e^{ikx}$$

- Thus:

$$\frac{df_k(t)}{dt} = (ik)^n f_k(t) = \sigma f_k(t) \quad \text{for } \sigma = (ik)^n$$

- And: $f_k(t) = f_k(0) e^{\sigma t}, \quad f(x, t) = \sum_{k=-\infty}^{\infty} f_k(0) e^{ikx + \sigma t}$

– “Phase speed”: $c = -\sigma / ik$

Fourier (Error) Analysis: Generic equation

- Generic PDE, FT:

$$f(x,t) = \sum_{k=-\infty}^{\infty} f_k(0) e^{ikx + \sigma t}$$

$$\frac{d f_k(t)}{dt} = \sigma f_k(t) \quad \text{for } \sigma = (ik)^n$$

$$n = 2q \quad \Rightarrow \quad (ik)^n = (-1)^q k^{2q} \quad (\text{real})$$

$$n = 2q - 1 \quad \Rightarrow \quad (ik)^n = -i (-1)^q k^{2q-1} \quad (\text{imaginary})$$

- Hence:

$n = 1$	$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x}$	$\sigma = ik$	Propagation: $c = -\sigma / ik = -1$, No dispersion
$n = 2$	$\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}$	$\sigma = -k^2$	Decay
$n = 3$	$\frac{\partial f}{\partial t} = \frac{\partial^3 f}{\partial x^3}$	$\sigma = -ik^3$	Propagation: $c = -\sigma / ik = +k^2$, With dispersion
$n = 4$	$\frac{\partial f}{\partial t} = \pm \frac{\partial^4 f}{\partial x^4}$	$\sigma = \pm k^4$	$+$: (Fast) Growth, $-$: (Fast) Decay

- Etc

Fourier Error Analysis: 1st derivatives $\frac{\partial f}{\partial x}$

- In the decomposition: $f(x, t) = \sum_{k=-\infty}^{\infty} f_k(t) e^{ikx}$
 - All components are of the form: $f_k(t) e^{ikx}$
 - Exact 1st order spatial derivative: $\frac{\partial f_k(t) e^{ikx}}{\partial x} = f_k(t) ik e^{ikx} = f_k(t) (ik e^{ikx})$
 - However, if we apply the centered finite-difference (2nd order accurate):

$$\left(\frac{\partial f}{\partial x} \right)_j = \frac{f_{j+1} - f_{j-1}}{2\Delta x} \Rightarrow$$

$$\left(\frac{\partial e^{ikx}}{\partial x} \right)_j = \frac{e^{ik(x_j + \Delta x)} - e^{ik(x_j - \Delta x)}}{2\Delta x} = \frac{(e^{ik\Delta x} - e^{-ik\Delta x}) e^{ikx_j}}{2\Delta x} = i \frac{\sin(k\Delta x)}{\Delta x} e^{ikx_j} = i k_{\text{eff}} e^{ikx_j}$$

where $k_{\text{eff}} = \frac{\sin(k\Delta x)}{\Delta x}$ (uniform grid resolution Δx)

- k_{eff} = effective wavenumber
- For low wavenumbers (smooth functions): $k_{\text{eff}} = \frac{\sin(k\Delta x)}{\Delta x} = k - \frac{k^3 \Delta x^2}{6} + \dots$
 - Shows the 2nd order nature of center-difference approx. (here, of k by k_{eff})

Fourier Error Analysis, Cont'd:

Effective Wave numbers

- Different approximations $\left(\frac{\partial e^{ikx}}{\partial x}\right)_j$ have different effective wavenumbers

- CDS, 2nd order: $k_{\text{eff}} = \frac{\sin(k\Delta x)}{\Delta x} = k - \frac{k^3 \Delta x^2}{6} + \dots$

- CDS, 4th order: $k_{\text{eff}} = \frac{\sin(k\Delta x)}{3\Delta x} (4 - \cos(k\Delta x))$

- Padé scheme, 4th order: $i k_{\text{eff}} = \frac{3i \sin(k\Delta x)}{(2 + \cos(k\Delta x)) \Delta x}$

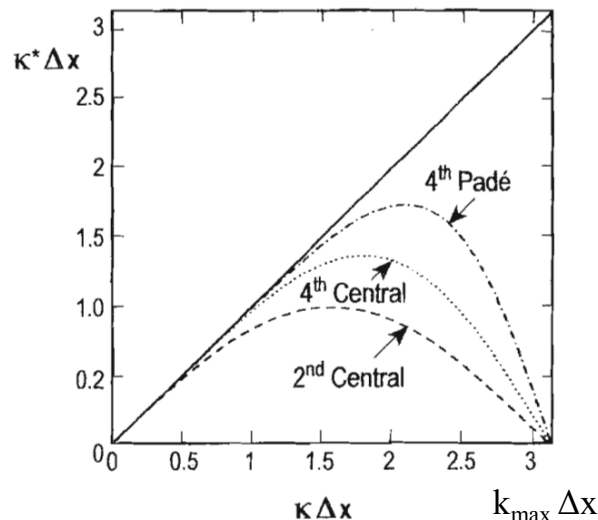


Fig. 3.4. Modified wavenumber for various schemes

The fourth-order Padé scheme is given by

$$(\delta_x u)_{j-1} + 4(\delta_x u)_j + (\delta_x u)_{j+1} = \frac{3}{\Delta x} (u_{j+1} - u_{j-1}) .$$

The modified wavenumber for this scheme satisfies⁶

$$i\kappa^* e^{-i\kappa\Delta x} + 4i\kappa^* + i\kappa^* e^{i\kappa\Delta x} = \frac{3}{\Delta x} (e^{i\kappa\Delta x} - e^{-i\kappa\Delta x}) ,$$

which gives

$$i\kappa^* = \frac{3i \sin \kappa\Delta x}{(2 + \cos \kappa\Delta x) \Delta x} .$$

Note that k_{eff} is bounded: $0 \leq k_{\text{eff}} \leq k_{\text{max}}$

$$k_{\text{max}} = \frac{\pi}{\Delta x}$$

Fourier Error Analysis, Cont'd

Effective Wave Speeds

Different approximations $\left(\frac{\partial e^{ikx}}{\partial x} \right)_j$ also lead to different effective wave speeds:

- Consider linear convection equations: $\frac{\partial f}{\partial t} + c \frac{\partial f}{\partial x} = 0$
 - For the exact solution: $f(x, t) = \sum_{k=-\infty}^{\infty} f_k(0) e^{ikx + \sigma t} = \sum_{k=-\infty}^{\infty} f_k(0) e^{ik(x - ct)}$ (since $\sigma = -ikc$)
 - For the numerical sol.: if $f = f_k^{num.}(t) e^{ikx} \Rightarrow \frac{df_k^{num.}}{dt} e^{ikx_j} = -f_k^{num.}(t) c \left(\frac{\partial e^{ikx}}{\partial x} \right)_j = -f_k^{num.}(t) c (ik_{eff} e^{ikx_j})$

which we can solve exactly (our interest here is only error due to spatial approx.)

$$\Rightarrow f_k^{num.}(t) = f_k(0) e^{-ik_{eff} c t}$$

$$\Rightarrow f^{numerical}(x, t) = \sum_{k=-\infty}^{\infty} f_k(0) e^{ikx - ik_{eff} c t} = \sum_{k=-\infty}^{\infty} f_k(0) e^{ik(x - c_{eff} t)}$$

$$\Rightarrow \frac{c_{eff}}{c} = \frac{\sigma_{eff}}{\sigma} = \frac{k_{eff}}{k} \quad (\text{defining } \sigma_{eff} = -ik_{eff} c = -ik c_{eff})$$

- Often, $c_{eff}/c < 1 \Rightarrow$ numerical solution is too slow.
- Since c_{eff} is a function of the effective wavenumber the scheme is dispersive (even though the PDE is not)

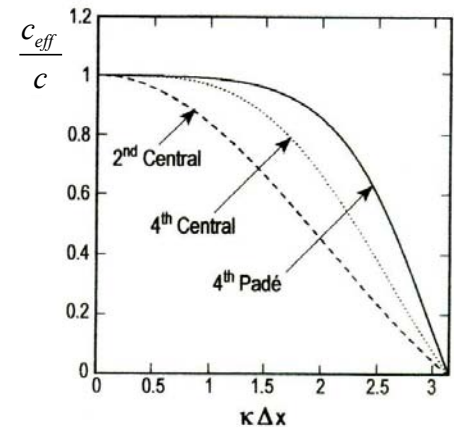


Fig. 3.5. Numerical phase speed for various schemes

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Source: Lomax, H., T. Pulliam, D. Zingg. *Fundamentals of Computational Fluid Dynamics*. Springer, 2001.

Evaluation of the Stability of a FD Scheme:

Three main approaches

Recall: $\tau_{\Delta x} = \mathcal{L}(\phi) - \hat{\mathcal{L}}_{\Delta x}(\hat{\phi} + \varepsilon) = -\hat{\mathcal{L}}_{\Delta x}(\varepsilon)$ Stability: $\|\hat{\mathcal{L}}_{\Delta x}^{-1}\| < \text{Const.}$ (for linear systems)

- **Heuristic stability:**

- Stability is defined with reference to an error (e.g. round-off) made in the calculation, which is damped (stability) or grows (instability)
- Heuristic Procedure: Try it out
 - Introduce an isolated error and observe how the error behaves
 - Requires an exhaustive search to ensure full stability, hence mainly informational approach

- **Energy Method**

- Basic idea:
 - Find a quantity, L_2 norm e.g. $\sum_j (\phi_j^n)^2$
 - Shows that it remains bounded for all n
- Less used than Von Neumann method, but can be applied to nonlinear equations and to non-periodic BCs

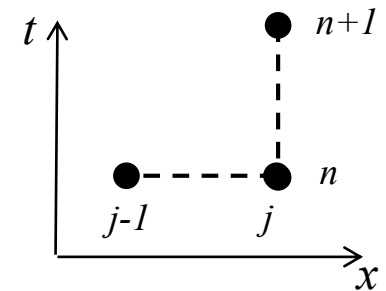
- **Von Neumann method (Fourier Analysis method)**

Evaluation of the Stability of a FD Scheme

Energy Method Example

- Consider again: $\frac{\partial \phi}{\partial t} + c \frac{\partial \phi}{\partial x} = 0$
- A possible FD formula (“upwind” scheme for $c > 0$): $\frac{\phi_j^{n+1} - \phi_j^n}{\Delta t} + c \frac{\phi_j^n - \phi_{j-1}^n}{\Delta x} = 0$
 $(t = n\Delta t, x = j\Delta x)$ which can be rewritten:

$$\phi_j^{n+1} = (1 - \mu) \phi_j^n + \mu \phi_{j-1}^n \quad \text{with} \quad \mu = \frac{c \Delta t}{\Delta x}$$



Derivation removed due to copyright restrictions. For the rest of this derivation, please see equations 2.18 through 2.22 in Durran, D. *Numerical Methods for Wave Equations in Geophysical Fluid Dynamics*. Springer, 1998. ISBN: 9780387983769.

Evaluation of the Stability of a FD Scheme

Energy Method Example

Derivation removed due to copyright restrictions. For the rest of this derivation, please see equations 2.18 through 2.22 in Durran, D. *Numerical Methods for Wave Equations in Geophysical Fluid Dynamics*. Springer, 1998. ISBN: 9780387983769.

Von Neumann Stability

dely used procedure

- Assumes initial error can be represented as a Fourier Series and considers growth or decay of these errors
- In theoretical sense, applies only to periodic BC problems and to linear problems
 - Superposition of Fourier modes can then be used

• Again, use,

but for the error:

$$\varepsilon(x, t) = \sum_{\beta=-\infty}^{\infty} \varepsilon_{\beta}(t) e^{i\beta x}$$

- Being interested in error growth/decay, consider only one mode:

$$\underline{\varepsilon_{\beta}(t) e^{i\beta x} \approx e^{\gamma t} e^{i\beta x}} \quad \text{where } \gamma \text{ is in general complex and function of } \beta : \gamma = \gamma(\beta)$$

- Strict Stability: The error will not to grow in time if

$$|e^{\gamma t}| \leq 1 \quad \forall \gamma$$

– in other words, for $t = n\Delta t$, the condition for strict stability can be written:

$$\underline{|e^{\gamma \Delta t}| \leq 1} \quad \text{or for } \underline{\xi = e^{\gamma \Delta t}}, \quad \underline{|\xi| \leq 1} \quad \text{von Neumann condition}$$

Norm of amplification factor ξ smaller or equal to 1

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